Lab Report No _6_

Introduction to Keil Debugger and simulator with Led Display Control using MTS-51.

Embedded Systems

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Lab Task No 6:

Solution:

Brief description (3-5 lines)
Write a program which Turn on LEDs according to the following cases and repeat each case 8 times. a. Case 1: Turn on LEDs from left to right. b. Case 1: Turn on LEDs from right to left. c. Blink two sets (left and right sides) of LEDs alternately. d. Blink all of 8 LEDs.
The code

```
ORG 0000H
START:
   MOV P1, #00H
   MOV R0, #08H
CASE1_LEFT:
   MOV R1, #00H
LEFT_LOOP:
   MOV P1, #00H
   SETB P1.0
   ACALL DELAY
   INC R1
   CJNE R1, #08H, LEFT_LOOP
   ACALL DELAY
CASE2_RIGHT:
   MOV R1, #07H
RIGHT_LOOP:
   MOV P1, #00H
   SETB P1.7
   ACALL DELAY
   DJNZ R1, RIGHT_LOOP
   ACALL DELAY
CASE3_BLINK_SETS:
```

```
BLINK_LOOP:
   MOV P1, #F0H
   ACALL DELAY
   MOV P1, #0FH
   ACALL DELAY
   DJNZ R0, BLINK_LOOP
CASE4_BLINK_ALL:
  BLINK_ALL_LOOP:
   MOV P1, #00H
   ACALL DELAY
   MOV P1, #FFH
   ACALL DELAY
   DJNZ R0, BLINK_ALL_LOOP
JMP START
DELAY:
   MOV R2, #255
DELAY_OUTER:
   MOV R3, #255
DELAY_INNER:
   DJNZ R3, DELAY_INNER
   DJNZ R2, DELAY_OUTER
   RET
END
```

The results (Screenshot)

Build Output

```
Program Size: data=9.0 xdata=0 code=81
".\Objects\8 LEDs" - 0 Error(s), 3 Warning(s).
Build Time Elapsed:  00:00:02
```

Build Output

```
creating hex file from ".\Objects\8 LEDs"...
".\Objects\8 LEDs" - 0 Error(s), 3 Warning(s).
Build Time Elapsed:  00:00:00
```

Conclusion

In conclusion, the lab on "Introduction to Keil Debugger and Simulator with LED Display Control using MTS-51" provided hands-on experience in using Keil's debugging tools and simulator capabilities.

By controlling an LED display, participants practiced monitoring program execution, examining memory content, and adjusting breakpoints for error tracking, reinforcing skills in real-time debugging and enhancing proficiency in embedded system development.